6
Electric fields

The effect of an electric field on Rydberg atoms, the Stark effect, provides an interesting example of how the deviation from the coulomb potential at small r in, for example an alkali atom, significantly alters its behavior in a field from hydrogenic behavior. The difference is somewhat surprising since field effects are fundamentally long range effects, and this difference alone would make the Stark effect worthy of study. However, in addition to its intrinsic interest, the Stark effect is of great practical importance for the study of Rydberg atoms.

Hydrogen

We begin by considering the behavior of the H atom in a static field, ignoring the spin of the electron. We start with the familiar zero field $n\ell m$ angular momentum eigenstates with the immediate goal of showing that important qualitative features of the Stark effect are easily understood using familiar notions. If the applied field E is in the z direction, the potential seen by the electron is given by

$$V = -\frac{1}{r} + Ez. \tag{6.1}$$

Using the zero field $n\ell m$ states we calculate the matrix elements $\langle n\ell \mathrm{m}|\mathrm{Ez}|\mathrm{n}'\ell' m'\rangle$ of the Stark perturbation to the zero field Hamiltonian. Writing the matrix element in spherical coordinates and choosing the z axis as the axis of quantization,

$$\langle n\ell m|Ez|n'\ell' m'\rangle = E\langle n\ell m|r\cos\theta|n'\ell' m'\rangle. \tag{6.2}$$

From the properties of the spherical harmonic angular functions we know that the Ez matrix elements are nonvanishing if $m' = m$ and $\ell' = \ell \pm 1$. The most obvious effect of the field is that it lifts the degeneracy of the ℓm states of a particular n. If we neglect the electric dipole coupling to other n states, the Hamiltonian matrix has the same entry, $-1/2n^2$, for all the diagonal elements and a set of off diagonal $\langle\ell|Ez|\ell\pm1\rangle$ matrix elements, all of which are proportional to E. If we subtract the common energy $-1/2n^2$ we find that all entries of the matrix are now proportional to E. Therefore, when the matrix is diagonalized to find its eigenvalues, they must all be proportional to E, i.e. the eigenvalues are given by $\lambda_i E$. Stated another way, the H atom exhibits a linear Stark shift. Since the eigenvalues are all proportional to E, the eigenvectors are independent of E, and the Stark states are thus field independent linear combinations of zero field ℓ states of the same n and m. The fact that the Stark states have linear Stark shifts means that they have permanent electric dipole moments. Stark states which have positive Stark shifts have the electron localized on the upfield side of the atom and Stark states which have negative Stark shifts have the electron localized on the downfield side of the atom.

Since m is a good quantum number, each set of m states is independent of the others. We consider first the circular $|m| = n - 1$ states. They have no first order Stark shift since there are no other states of the same m and n. There are, however, two $m=n-2$ states, which exhibit small linear Stark shifts. The Stark shifts are not large since the radial matrix element is small,[1]

$$\langle n\ell|r|n\ell+1\rangle = \frac{-3n\sqrt{n^2-\ell^2}}{2}. \tag{6.3}$$

The minus sign is for radial wavefunctions $R_{n\ell}(r) \sim +r^{\ell}$ as $r \rightarrow 0$. On the other hand, the $m = 0$ states also include the low ℓ states for which the radial matrix element is large, and the extreme $m = 0$ states have Stark shifts of approximately $\pm 3n^2E/2$.

The most straightforward way to treat the Stark effect is to use parabolic coordinates, for in parabolic coordinates the problem remains separable even with the electric field.[1–3] The parabolic coordinates are defined in terms of the familiar Cartesian and spherical coordinates by

$$\begin{aligned} \xi &= r + z = r(1+\cos\theta) \\ \eta &= r - z = r(1-\cos\theta) \\ \phi &= \tan^{-1} y/x, \end{aligned} \tag{6.4}$$

and

$$\begin{aligned} x &= \sqrt{\xi\eta}\cos\phi \\ y &= \sqrt{\xi\eta}\sin\phi \\ z &= (\xi-\eta)/2 \\ r &= (\xi+\eta)/2. \end{aligned} \tag{6.5}$$

Surfaces of constant ξ or η are paraboloids of revolution about the z axis. From Eqs. (6.4) it is apparent that $\xi = 0$ corresponds to the $-z$ axis, and $\xi = \infty$ corresponds to $r \rightarrow \infty$, $\theta \neq 0$. Correspondingly, $\eta = 0$ corresponds to the $+z$ axis and $\eta \rightarrow \infty$ corresponds to $r \rightarrow \infty$, $\theta \neq 0$. When the field is pointing in the z direction, corresponding to Eq. (6.1), the electron escapes to $\eta = \infty$. Equivalently, the motion in the ξ direction is bound, but the motion in the η direction is not.

In parabolic coordinates the Schroedinger equation for an electron orbiting a singly charged ion with an external field in the z direction, i.e. in the potential given by Eq. (6.1), is written using Eqs. (6.5) as[1–3]

$$\left[-\frac{\nabla^2}{2}-\frac{2}{\xi+\eta}+\frac{E(\xi-\eta)}{2}\right]\psi = W\psi, \tag{6.6}$$

where

$$\nabla^2 = \frac{4}{\xi+\eta}\frac{\partial}{\partial\xi}\left(\xi\frac{\partial}{\partial\xi}\right)+\frac{4}{\xi+\eta}\frac{\partial}{\partial\eta}\left(\eta\frac{\partial}{\partial\eta}\right)+\frac{1}{\xi\eta}\frac{\partial^2}{\partial\phi^2},$$

and W is the energy.

If we assume that solution can be written as the product,[1–3]

$$\Psi(\xi,\eta,\phi) = u_1(\xi)u_2(\eta)\mathrm{e}^{im\phi}, \tag{6.7}$$

with m an integer or zero, and substitute Eq. (6.7) into Eq. (6.6), we find two independent equations for $u_1(\xi)$ and $u_2(\eta)$

$$\frac{\mathrm{d}}{\mathrm{d}\xi}\left(\xi\frac{\mathrm{d}u_1}{\mathrm{d}\xi}\right)+\left(\frac{W\xi}{2}+Z_1-\frac{m^2}{4\xi}-\frac{E\xi^2}{4}\right)u_1 = 0 \tag{6.8a}$$

and

$$\frac{\partial}{\mathrm{d}\eta}\left(\eta\frac{\mathrm{d}u_2}{\mathrm{d}\eta}\right)+\left(\frac{W\eta}{2}+Z_2-\frac{m^2}{4\eta}+\frac{E\eta^2}{4}\right)u_2 = 0. \tag{6.8b}$$

From Eqs. (6.8a) and (6.8b) it is apparent that the sign of m is unimportant, and that the wavefunctions for $\pm m$ are degenerate, as might be expected for this cylindrically symmetric problem. In Eqs. (6.8a) and (6.8b) the separation constants Z_1 and Z_2 are related by[1,2]

$$Z_1 + Z_2 = 1. \tag{6.9}$$

Z_1 and Z_2 may be thought of as the positive charges binding the electron in the ξ and η coordinates. This point will become more apparent shortly. Note that the two differences between Eqs. (6.8a) and (6.8b) are that the field enters with different signs in the two equations and that Z_1 and Z_2 may have different values.

The classic way of determining the energies of hydrogenic levels in a field is to solve the zero field problem in parabolic coordinates and calculate the effect of the field using perturbation theory. The zero field parabolic wavefunctions obtained by solving Eqs. (6.8a) and (6.8b) have, in addition to the quantum numbers n and $|m|$, the parabolic quantum numbers n_1 and n_2, which are nonnegative integers.[1] n_1 and n_2 are the numbers of nodes in the u_1 and u_2 wavefunctions and are related to n and $|m|$ by

$$n = n_1 + n_2 + |m| + 1. \tag{6.10}$$

They are in addition related to the effective charges Z_1 and Z_2 by

$$Z_1 = \frac{1}{n}\left(n_1 + \frac{|m|+1}{2}\right) \tag{6.11a}$$

and

$$Z_2 = \frac{1}{n}\left(n_2 + \frac{|m|+1}{2}\right). \tag{6.11b}$$

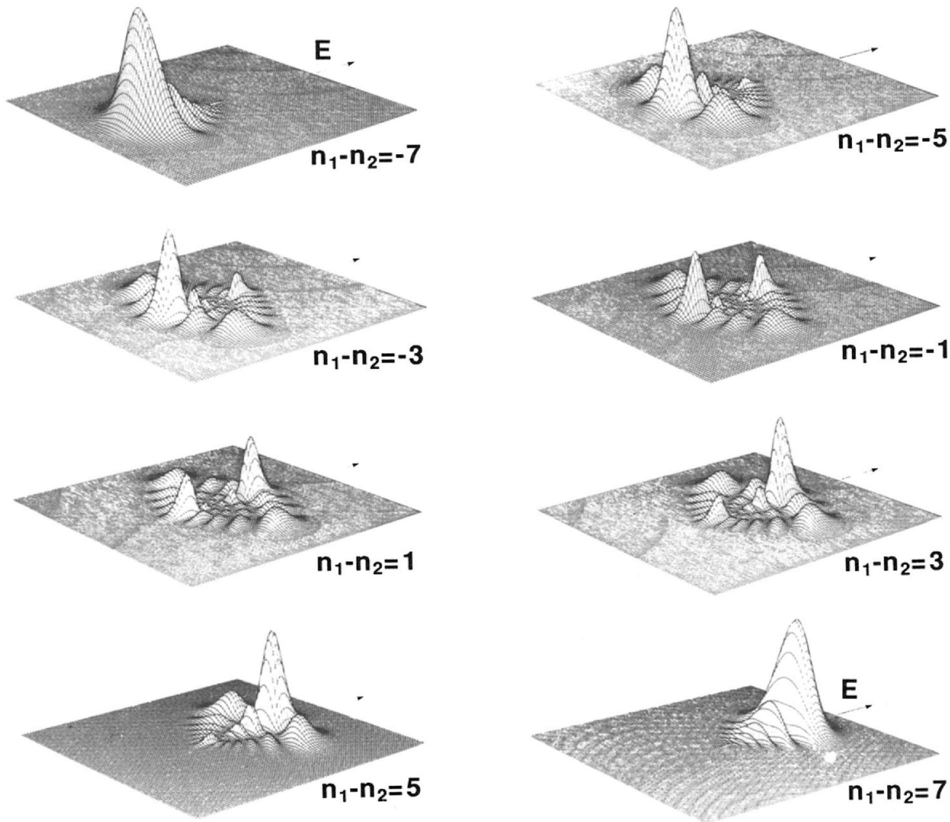

Fig. 6.1 Charge distribution for H, for "parabolic" eigenstates: $n = 8, m = 0, n_1 - n_2 = -7$ to 7. The dipole moments that give rise to the first order Stark effect are conspicuous (from ref. 4).

The wavefunctions are given in terms of associated Laguerre polynomials, and using the approximate form of the Laguerre polynomials for large arguments, we can write an approximate, unnormalized wavefunction as[1]

$$\psi_{nn_1n_2m} \propto e^{im\phi}\, \xi^{n_1+|m|/2}\, \eta^{n_2+\,|m|/2}\, e^{-(\xi+\eta)/2n}. \tag{6.12}$$

Taking the squared absolute value of Eq. (6.12) and using the definition of the parabolic coordinates given in Eq. (6.4), we can write an expression for the electron probability distribution in spherical coordinates,[1]

$$|\psi_{nn_1n_2m}|^2 = r^{2n-2}(1 + \cos\theta)^{2n_1+|m|}(1 - \cos\theta)^{2n_2+|m|}e^{-2r/n}. \tag{6.13}$$

Eq. (6.13) shows explicitly that the $n_1 - n_2 \approx n$ wavefunctions are localized along the $+z$ axis while the $n_2 - n_1 \approx n$ wavefunctions are localized along the $-z$ axis. Similarly, $n_1 - n_2 \approx 0$ wavefunctions are localized near the $z = 0$ plane. In Fig. 6.1 we show the $n = 8,\ m = 0,\ n_1 - n_2 = -7$ to 7 wavefunctions to illustrate the polarization along the field direction.[4] Careful inspection also reveals that the nodes lie on parabolas. Using the zero field wavefunctions the first order energies are[1]

$$W_{nn_1n_2m} = \frac{-1}{2n^2} + \frac{3}{2}E(n_1 - n_2)n. \tag{6.14}$$

While there is no explicit m dependence, there is an implicit m dependence due to the fact that $n_1 + n_2 + m + 1 = n$. For $m = 0$, allowed values of $n_1 - n_2$ are $n - 1$, $n - 3 \ldots, -n + 1$ while for $m = 1$ they are $n - 2, n - 4 \ldots -n + 2$. The even and odd m levels are interleaved. Furthermore, for the quantum numbers n and m there are $n - |m|$ Stark levels. Note that for the circular states, $|m| = n - 1, n_1 = n_2 = 0$, and the first order Stark shift vanishes, as we have already seen.

By taking into account the matrix elements off diagonal in n, the second and higher order contributions to the Stark effect can be calculated. If the calculation is carried through second order, the energies are given by[1]

$$W_{nn_1n_2m} = \frac{-1}{2n^2} + \frac{3\,En}{2}(n_1 - n_2) - \frac{E^2}{16}n^4\,[17n^2 - 3(n_1 - n_2)^2 - 9m^2 + 19]. \tag{6.15}$$

The second order shift is always to lower energy, as expected from oscillator strength sum rules.[5] Equally important, the second order shift breaks the m degeneracy. The energies can be calculated to higher order,[6,7] but for many applications the first and second order shifts are adequate. This point is made explicitly by Fig. 6.2, a plot of the H $|m| = 1$ energy levels from $n = 8$ to $n=14$ in electric fields from 0 to 2×10^{-5} (0 to 10^5 V/cm). Fig. 6.2 makes several important points about the Stark effect in H. First, the levels exhibit apparently linear Stark shifts from zero field to the point at which field ionization occurs at significant rates, shown by the broken lines in Fig. 6.2. Only when looking along the diagram from the side of the page is the curvature of the levels evident. Second, the Stark levels of adjacent n cross; there is no coupling, at least at this level of resolution. Finally, the red, or down shifted, Stark states ionize near the classical ionization limit, but the higher energy blue, or upshifted, states ionize only at higher fields.

As shown by Fig. 6.2, the Stark shifts are quite linear, except at the highest fields shown, and the first order energies of Eq. (6.14) are adequate for many purposes. On the other hand, even the second order energies of Eq. (6.15) are not adequate for comparisons to precise measurements of the energy levels in field. For example, using a fast H beam Koch[8] observed the (10,8,0,1) $\rightarrow$ (25,21,2,1) and (10,0,9,0) $\rightarrow$ (30,0,29,0) transitions in fields of 2.514 and 0.689 kV/cm respectively using the 10 μm R24 and P24 CO_2 laser lines. Here the states are denoted $(n,n_1,n_2,|m|)$. From his results it is apparent that the energy of the red shifted (30,0,29,0) state is given accurately by perturbation theory results at about tenth order. In contrast, the blue shifted (25,21,2,1) state is never given to the same accuracy, and, in fact, the perturbation theory result does not converge but oscillates about the experimental result with the minimum amplitude of the oscillations at about tenth order.[8]

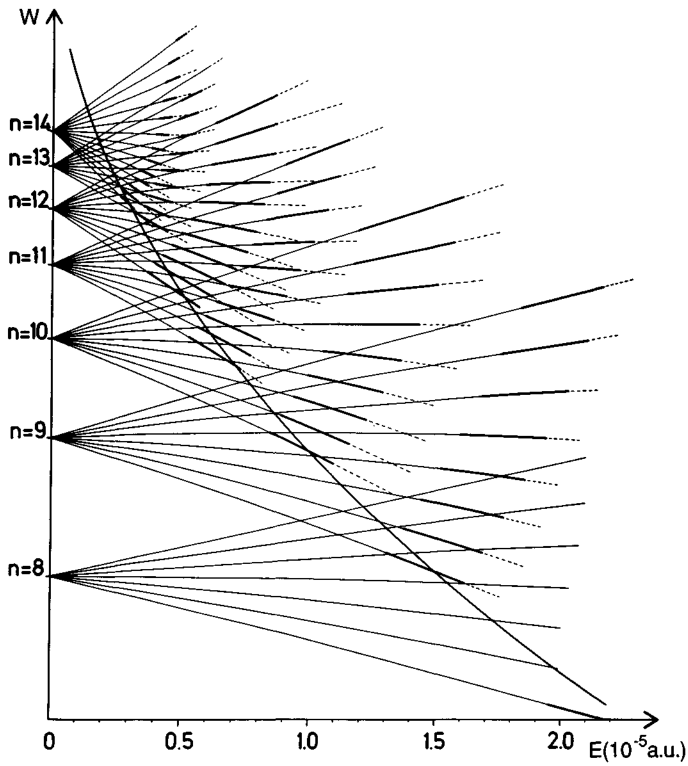

Fig. 6.2 Stark structure and field ionization properties of the $|m| = 1$ states of the H atom. The zero field manifolds are characterized by the principal quantum number n. Quasi-discrete states with lifetime $\tau > 10^{-6}$ s (solid line), field broadened states 5×10^{-10} s $< \tau < 5 \times 10^{-6}$ s (bold line), and field ionized states $\tau < 5 \times 10^{-10}$ s (broken line). Field broadened Stark states appear approximately only for $W > W_c$. The saddle point limit $W_c = -2\sqrt{E}$ is shown by a heavy curve (from ref. 3).

The second point made by Fig. 6.2, although not with very high resolution, is that the levels of n and $n + 1$ cross. The extreme $m = 0$ red and blue states have Stark shifts of approximately $\pm 3n^2E/2$, which combined with the n – (n + 1) energy spacing of $1/n^3$, yields a crossing field of

$$E = 1/3n^5. \tag{6.16}$$

This field is associated with the Inglis–Teller limit, where Stark broadening in a plasma causes levels of adjacent n to become unresolvable.[9] The fact that the red $n + 1$ and blue n levels actually cross, in spite of the fact that they have the same m, is a result of the fact that the use of the parabolic coordinates diagonalizes the Runge–Lenz vector, which has a different eigenvalue in the two crossing states. The fact that both the Runge–Lenz vector and angular momentum are conserved in H in zero field was exploited by Park[10] to show in an elegant way that the transformation coefficient between the parabolic nn_1n_2m states and the spherical

$n\ell m$ states was simply a Clebsch–Gordon coefficient. If we expand the Stark state $|nn_1n_2m\rangle$ as

$$|nn_1n_2m\rangle = \sum_{\ell} |\, n\ell m\rangle\langle n\ell m|nn_1n_2m\rangle\,, \tag{6.17}$$

then the transformation coefficient is given in terms of Wigner 3J symbols as

$$\langle nn_1n_2m|n\ell m\rangle = (-1)^{(1-n+m+n_1-n_2)/2\,+\,\ell} \times \sqrt{2\ell+1}\begin{pmatrix} \frac{n-1}{2} & \frac{n-1}{2} & \ell \\ \frac{m+n_1-n_2}{2} & \frac{m-n_1+n_2}{2} & -\,m \end{pmatrix}. \tag{6.18}$$

An equivalent form is given by Englefield.[11] It is possible to find quite a variety of phases for the transformation coefficients of Eq. (6.18).[10–13] The phase depends on the phase conventions established for the spherical and parabolic states. The choice of phase in Eq. (6.18) is for spherical functions with an r^{ℓ}, as opposed to $(-r)^{\ell}$, dependence at the origin and the spherical harmonic functions of Bethe and Salpeter. A few examples of the spherical harmonics are given in Table 2.2. The parabolic functions are assumed to have an $(\xi n)^{|m|/2}$ behavior at the origin and an $e^{im\phi}$ angular dependence. This convention means, for example, that for all Stark states with the quantum number m, the transformation coefficient $\langle nn_1n_2m|nmm\rangle$ is positive. To the extent that the Stark effect is linear, i.e. to the extent that the wavefunctions are the zero field parabolic wavefunctions, the transformation of Eqs. (6.17) and (6.18) allows us to decompose a parabolic Stark state in a field into its zero field components, or vice versa.

Harmin[14] derived an approximate form of the transformation of Eq. (6.18) which is particularly useful. The information contained in the quantum numbers n_1 and n_2 always appears as $\pm(n_1 - n_2)$ and can also be represented by Z_1 and Z_2, the separation constants, a notion which easily passes into the regime in which n_2 is not a well defined quantum number. (In a field n_1 is always a good quantum number.) Explicitly, Harmin showed that Eq. (6.18) can also be written as[14]

$$\langle nn_1n_2m|n\ell m\rangle = \sqrt{\frac{2}{n}}(-1)^{\ell}\, P_{\ell m}(Z_1 - Z_2), \tag{6.19}$$

where $P_{\ell m}$ is a normalized associated Legendre polynomial of Bethe and Salpeter.[1] It is defined for $m < 0$, and $\int_{-1}^{1} P^2_{\ell m}(x)\mathrm{d}x = 1$.[1] Eq. (6.19) is only defined over the region $-1 \leq Z_1 - Z_2 \leq 1$, which is, however, a less restrictive requirement than that both n_1 and n_2 quantum numbers have integral values. We may also write Eq. (6.19) in terms of the parabolic n_1 and n_2 quantum numbers using $Z_1 - Z_2 = (n_1 - n_2)/n$ from Eq. (6.11). From Eq. (6.19) it is easy to see how the parabolic states are composed of the $n\ell m$ states. From either the properties of the 3J symbol or the fact that $P_{00}(x) = 1/\sqrt{2}$ it is apparent that the ns state is evenly

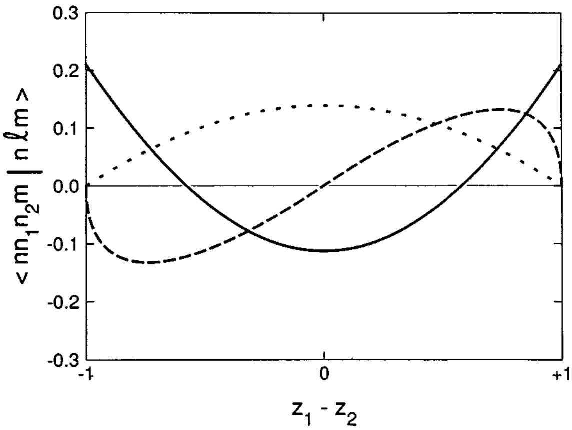

Fig. 6.3 Plots of the projection of the $n = 100$ Stark states of $m = 0, 1$, and 2 on the $n = 100$ zero field d states, <100 n_1n_2m|100 2m> for $m = 0, 1$, and 2: $m = 0$ (——), $m = 1$ (– – –) $m = 2$ (. . .). The values are obtained from Eqs. (6.11) and (6.19).

spread over the nn_1n_20 states. The $\ell \geq 1$ states are more interesting. In Fig. 6.3 we plot Eq. (6.19) for $\ell = 2$ $|m| = 0, 1$, and 2. From Fig. 6.3 it is clear that the $m = 0$ states at the edge of the Stark manifold, i.e. those for which $|n_1 - n_2| \sim n$, have much more d character than those in the center. On the other hand, the $|m| = 2$ states at the center of the Stark manifold have substantial d character while the states at the edges do not. For $n = 10$ and $|m| = 2$ the largest value of $|Z_1 - Z_2| = |n_1 - n_2|/n$ is 0.7, but at higher n it approaches 1, at which point the $|m|=2$ Stark states at the edge of a Stark manifold have vanishingly small d character.

Working out the parabolic wavefunctions in terms of the Laguerre polynomials is useful in the analytic calculation of the Stark effect using perturbation theory. However, it is not useful in very strong electric fields. Here we outline a more general procedure which is valid in strong fields and lends itself to numerical computations. If we replace $u_1(\xi)$ and $u_2(\eta)$ in Eqs. (6.8a) and (6.8b) by[1–3,15]

$$u_1(\xi) = \frac{\chi_1(\xi)}{\sqrt{\xi}} \tag{6.20a}$$

$$u_2(\eta) = \frac{\chi_2(\eta)}{\sqrt{\eta}} \tag{6.20b}$$

and introduce a normalizing factor of $\sqrt{2\pi}$ to conform to common usage, the wavefunction is given by

$$\psi(\xi, \eta, m) = \frac{\chi_1(\xi)\,\chi_2(\eta)e^{im\phi}}{\sqrt{2\pi\varepsilon\eta}}, \tag{6.21}$$

which leads to the separated equations written as

$$\frac{d^2\chi_1}{d\xi^2} + \frac{\chi_1}{2}[W - V(\xi)] = 0 \tag{6.22a}$$

and

$$\frac{d^2\chi_2}{d\eta^2} + \frac{\chi_2}{2}[W - V(\eta)] = 0, \tag{6.22b}$$

where

$$V(\xi) = 2\left(-\frac{Z_1}{\xi} + \frac{m^2-1}{4\xi^2} + \frac{E\xi}{4}\right), \tag{6.23a}$$

and

$$V(\eta) = 2\left(-\frac{Z_2}{\eta} + \frac{m^2-1}{4\eta^2} - \frac{E\eta}{4}\right) \tag{6.23b}$$

The χ_1 and χ_2 wavefunctions describe the motion of particles of total energy $W/4$ in the potentials $V(\xi)/4$ or $V(\eta)/4$. It is also interesting to note that for the case $E=0$ the wave equations of Eqs. (6.22a) and (6.22b) are similar to the radial equation for the coulomb potential in spherical coordinates. Explicitly, making the substitutions

$$\begin{gathered} 2W \rightarrow W/2 \\ \ell(\ell+1) \rightarrow m^2 - 1 \\ Z \rightarrow Z_1, Z_2 \\ r \rightarrow \xi, \eta \end{gathered} \tag{6.24}$$

in Eq. (2.10) for $\rho(r) = rR(r)$ yields Eqs. (6.22a) and (6.22b).

When $E = 0$, the potential $V(\eta)$ with Z_2 substituted for Z_1 is the same as $V(\xi)$. It is apparent that for small values of Z_1 and Z_2 the potentials $V(\xi)$ and $V(\eta)$ are not as deep as for the higher values of Z_1 and Z_2. Thus at the same energy W the number of nodes which occur in the wavefunction increases with Z_1 or Z_2. Inspecting Eq. (6.24), we can see that when $E \neq 0$ that $V(\xi)$ and $V(\eta)$ are no longer identical for the same values of Z_1 and Z_2. The motion in the ξ direction is bounded for all energies since $V(\xi) \rightarrow E\xi$ as $\xi \rightarrow \infty$. In contrast, the motion in the η direction is unbounded, for any energy, since $V(\eta) \rightarrow -E\eta$ as $\eta \rightarrow \infty$. The different behaviors of $V(\xi)$ and $V(\eta)$ as ξ and $\eta \rightarrow \infty$ lead to qualitatively different wavefunctions. The motion in the ξ direction has a well defined integer quantum number, n_1, and the motion in the η direction is, in principle, a continuum, containing resonances. In practice, at energies far below the saddle point in $V(\eta)$ the motion in the η direction also has a good quantum number, n_2. In Fig. 6.4 we show the potentials $V(\xi)$ and $V(\eta)$ for $E=10^{-6}$, and $|m| = 1$. The potentials are shown for Z_1 and $Z_2 = 0.1$, 0.5, and 0.9, corresponding to a blue shifted Stark state, a middle state, and a red shifted state. As shown by Fig. 6.4(a), the ξ motion is always bounded. In Fig. 6.4(b) it is apparent that the saddle point in $V(\eta)$ occurs at lower energy for the red state ($Z_2 = 0.9$) than for the blue state. Had we drawn the potential $V(\eta)$ for $m=0$, it would be deeper at $\eta = 0$ and would lie generally lower than the $|m| = 1$ potential. For $|m| > 1$ the potential has a $1/\eta^2$ centrifugal barrier at small η and lies generally above the $m = 1$ potentials of Fig. 6.4.

To solve Eqs. (6.22a) and (6.22b) for $E \neq 0$ we take advantage of the fact that the motion in ξ is bounded. This motion always has a bound wavefunction and a

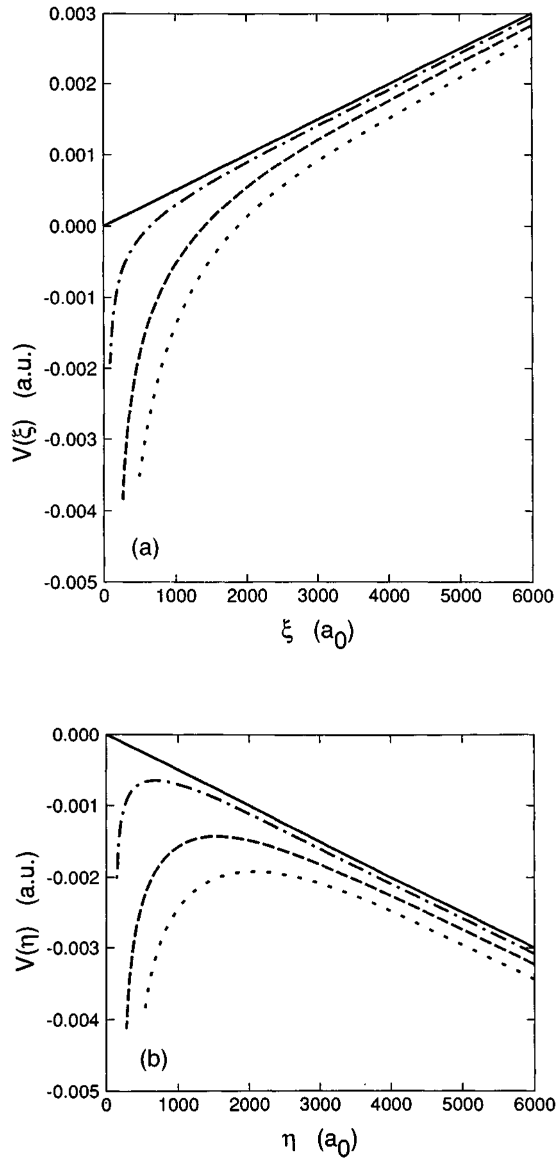

Fig. 6.4 Plots of the potentials $V(\xi)$ and $V(\eta)$ in a field $E = 10^{-6}$ for $|m| = 1$. (a) $V(\xi)$ for $Z_1 = 0$ (——), $Z_1 = 0.1$ (- - -), $Z_1 = 0.5$ (— — —), and $Z_1 = 0.9$ (· · ·) (b) $V(\eta)$ for $Z_2 = 0$ (——), $Z_2 = 0.1$ (- - -), $Z_2 = 0.5$(— — —), and $Z_2 = 0.9$ (· · ·). Note that the saddle point in $V(\eta)$ occurs at lower energy for larger values of Z_2. As $\eta \rightarrow \infty$ the Stark effect dominates both potentials, so that the ξ motion is always bounded.

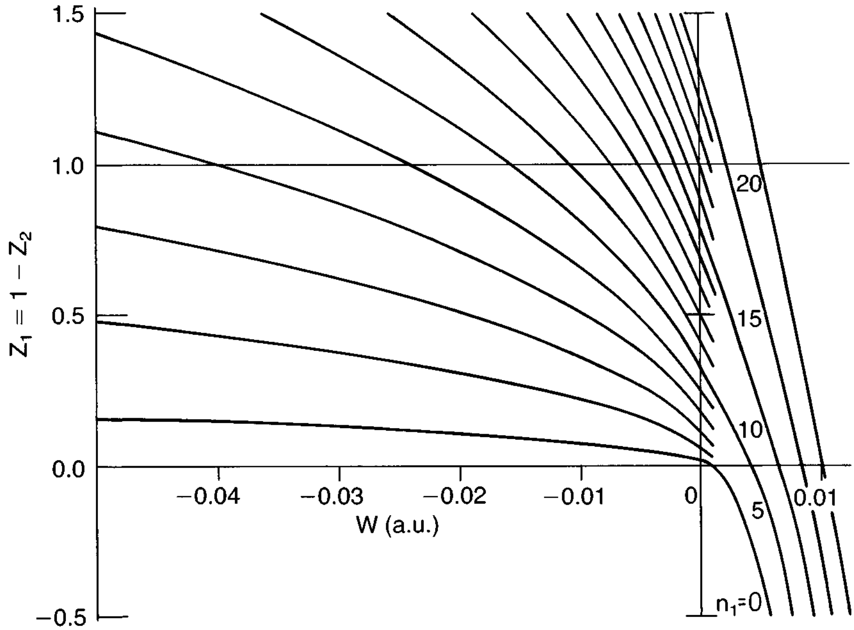

Fig. 6.5 Eigenvalues $Z_1 = 1 - Z_2$ as a function of energy for $m = 0$, $E = 1.5 \times 10^{-5}$ au = 77 kV/cm, and n_1 = 0–20 (from ref. 15).

well defined, integral quantum number, n_1, which is simply the number of nodes in the $\chi_1(\xi)$ wavefunction. For any energy W and any value of n_1 we can find the separation constant, or eigenvalue, Z_1. A straightforward way of finding the eigenvalues of Z_1 is to use the unnormalized WKB approximation i.e.[1,3,15]

$$\chi_1(\xi) = \frac{1}{[T(\xi)]^{1/4}} \cos\left[\int_{\xi_1}^{\xi} \sqrt{T(x)}\, dx - \frac{\pi}{4}\right], \tag{6.25}$$

where $T(\xi) = [W - V(\xi)]/2$, and ξ_1 is the inner turning point. The WKB wavefunction of Eq. (6.25) satisfies the quantization condition

$$\int_{\xi_l}^{\xi_2} \sqrt{T(\xi)}\, d\xi = \left(n_1 + \frac{1}{2}\right)\pi, \tag{6.26}$$

where n_1 is a positive integer and ξ_1 and ξ_2 are the inner and outer classical turning points, where $W = V(\xi)$. An alternative approach is use a Numerov algorithm to find the allowed wavefunctions, as discussed by Luc-Koenig and Bachelier.[3] In either case it is important to remember that for the same values of W, m, and E there exist many solutions to Eq. (6.26) corresponding to different values of the quantum number n_1 and the eigenvalue Z_1. These solutions are orthogonal, which is why the H Stark energy levels of the same m cross, as shown by Fig. 6.2.

In Fig. 6.5 we show the eigenvalues Z_1 associated with several values of the quantum number n_1 as a function of the energy W. These values of Z_1 are obtained by means of the WKB quantization condition of Eq. (6.26). For a constant value of n_1, the value of Z_1 decreases with increasing energy. Keeping n_1 constant means the WKB integral of Eq. (6.26) must remain a constant. The WKB integral is an integral of the momentum, or the square root of twice the kinetic energy, over the classically allowed spatial region. Increasing the energy increases both the spatial range covered by the integral and the kinetic energy. To keep the same value of n_1 as W is increased, the depth of the small ξ part of the potential is reduced, by reducing Z_1.

For fixed W, m, and E there exists a series of values of n_1 and Z_1 implying a series of allowed values of $Z_2 = 1 - Z_1$. The known value of Z_2 can now be used to solve Eq. (6.22b). Since $V(\eta) \to -E\eta/2$ as $\eta \to \infty$, in principle, the solution to Eq. (6.22b) is a continuum wave for all values of the energy. Accordingly we normalize the $\chi_2(\eta)$ wavefunction per unit energy as $\eta \to \infty$. The asymptotic form of the χ_2 wavefunction is easily obtained using the WKB approximation as[3]

$$\chi_2(\eta) = \frac{N}{(E\eta + 2W)^{1/4}} \sin\left[\frac{1}{3E}(E\eta + 2W)^{3/2} + \alpha\right], \tag{6.27}$$

where α is a phase. In Eq. (6.27) we see the expected oscillatory behavior and an amplitude decreasing as the square root of the momentum. Normalization per unit energy is often obtained by requiring that the inward or outward flux $I = 1/2\pi$. An approach consistent with the one given in Chapter 2 is to require

$$N^2 \int \psi^* \mathrm{d}\tau \int_{W-\Delta W}^{W+\Delta W} \psi \mathrm{d}W' = 1. \tag{6.28}$$

In parabolic coordinates the volume element $\mathrm{d}\tau = (\xi+\eta)\mathrm{d}\xi \mathrm{d}\eta \mathrm{d}\phi/4$. Using the general form of the wavefunction given in Eq. (6.21) and carrying out the angular integration, leads to

$$N^2 \int_0^\infty \int_0^\infty \chi_1^*(\xi)\, \chi_2^*(\eta) \left(\frac{1}{\eta} + \frac{1}{\xi}\right) \frac{\mathrm{d}\xi \mathrm{d}\eta}{4} \int_{W-\Delta W}^{W+\Delta W} \chi_1(\xi)\chi_2(\eta) \mathrm{d}W'. \tag{6.29}$$

The major contribution to the integral of Eq. (6.29) comes from $\eta \to \infty$, in which case the $1/\eta$ term may be ignored, leaving

$$N^2 \int_0^\infty \frac{(\chi_1^*(\xi)\chi_1(\xi))}{\xi} \mathrm{d}\xi \int_0^\infty \chi_2^*(\eta) \mathrm{d}\eta \int_{W-\Delta W}^{W+\Delta W} \chi_2(\eta) \mathrm{d}W' = 1. \tag{6.30}$$

Requiring that the bound $\chi_1(\xi)$ wavefunction be normalized according to

$$\int_0^\infty \frac{\chi_1^2(\xi)}{\xi} \mathrm{d}\xi = 1 \tag{6.31}$$

yields

$$N = \frac{1}{\sqrt{2\pi}}. \tag{6.32}$$

Thus, the amplitude of the continuum $\chi_2(\eta)$ wave varies insignificantly with energy as $r \to \infty$.

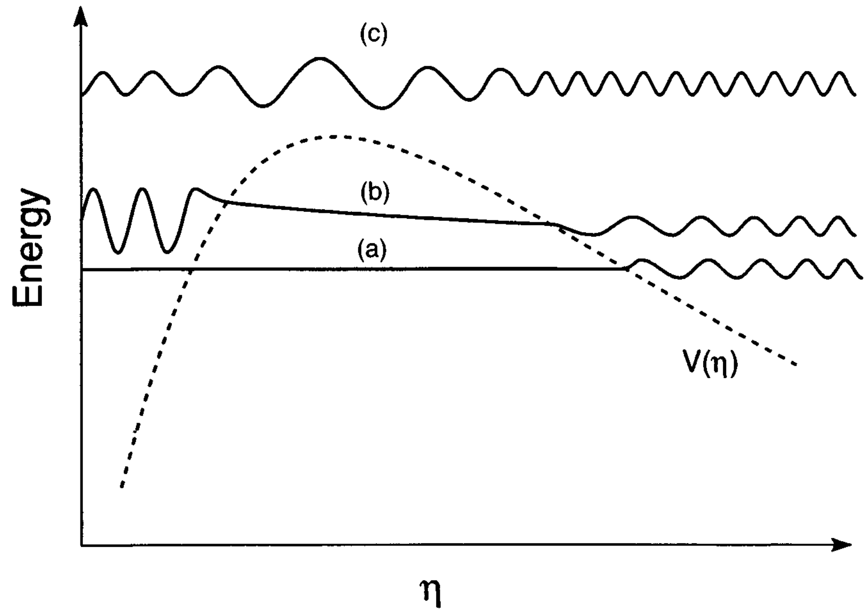

Fig. 6.6 Schematic wavefunctions $\chi_2(\eta)$ for several choices of energy W. (a) An energy below V_b the peak of the potential $V(\eta)$ but away from a resonance of the inner well of the potential. The wavefunction in vanishingly small in the inner well of the potential. (b) An energy below V_b and equal to one of the resonances of the inner well. (c) An energy above V_b.

We are not so much interested in the wavefunction at $r = \infty$ as we are in the wavefunction in the inner potential well of $V(\eta)$. Examining plots of the potential $V(\eta)$, it is apparent that there is, in general, a peak V_b in the potential for $\eta > 0$. If the energy W is above the peak of the potential, $W > V_b$, the entire region is classically accessible. If the energy is less than the peak of the potential barrier, V_b, there are two classically allowed regions, which are connected by a classically forbidden region through which tunnelling occurs. The ratio of the amplitude of the wavefunction in the inner well to that in the outer well is the quantity of interest.

Consider for a moment the energy dependence of $\chi_2(\eta)$ shown qualitatively in Fig. 6.6. Outside the barrier it has the standing wave form of Eq. (6.27) with virtually the same amplitude irrespective of energy. At an energy below the peak of the barrier the small transmission through the barrier allows some of the exterior wavefunction to penetrate into the inner well. Due to the small transmission, in general the wavefunction in the inner well of $V(\eta)$ is negligible, as shown by Fig. 6.6(a). However, at a resonance of the inner well, when an integral number, n_2, of nodes fit, the amplitude of the wavefunction in the inner well becomes very large, as shown in Fig. 6.6(b). This situation is no different from a microwave cavity tenuously coupled to a transmission line.

Far below the barrier in $V(\eta)$ the transmission through the barrier is vanishingly small and the resonances are so sharp as to be bound states for all practical purposes, i.e. n_2 is, for all intents and purposes, a good quantum number. As the energy is increased the transmission through the barrier increases, increasing the width of the resonances and decreasing the amplitude of $\chi_2(\eta)$ in the inner well. Finally, when the energy exceeds the top of the barrier, the wavefunction has an amplitude which is spatially smoothly varying, displaying minimal energy dependence as shown by Fig. 6.6(c). While Fig. 6.6 shows the variation of the amplitude of the wavefunction in the inner well of $V(\eta)$ in a qualitative way, we need a quantitative measure. A useful one takes advantage of the fact that for small values of ξ and η the regular solutions of $\chi_1(\xi)/\sqrt{\xi}$ and $\chi_2(\eta)/\sqrt{\eta}$ behave as $\xi^{|m|/2}$ and $\eta^{|m|/2}$. A useful approach is to define a small ξ, η wavefunction as[3]

$$\psi = \frac{\sqrt{C^m_{n_1}}}{\sqrt{2\pi}} (\xi\eta)^{|m|/2} e^{im\phi}, \tag{6.33}$$

where $C^m_{n_1}$, the density of states, is a measure of the probability of finding the electron near the origin, which is of practical interest for photoexcitation from the ground state and for field ionization of nonhydrogenic atoms. $C^m_{n_1}$ is easily obtained from the normalized $\chi_1(\xi)$ and $\chi_2(\eta)$ wavefunctions.

Figure 6.7 is a plot of $\sqrt{C^m_{n_1}}$, the square root of the density of states for $m = 0$, $n_1 = 7$, and $E = 1.5 \times 10^{-5}$ au. In this case the maximum in the potential barrier of $V(\eta)$ is at $V_b = -0.00372$. Above V_b there is, in essence, a continuum. Below V_b there is a series of sharp states which each have the approximate quantum numbers n_2 shown, corresponding to n_2 nodes of the $\chi_2(\eta)$ wavefunction in the inner potential well of $V(\eta)$. For any m, each value if the quantum number n_1 has a spectrum of $\sqrt{C^m_{n_1}}$ analogous to the one shown in Fig. 6.7.

Field ionization

Field ionization, which is of great practical importance in the study of Rydberg atoms, has been introduced implicitly in Figs. 6.2, 6.6, and 6.7. We now consider it explicitly. The order of magnitude of the field required for ionization can be estimated using the method outlined in Chapter 1. The combined coulomb–Stark potential,

$$V = -1/r + Ez, \tag{6.34}$$

has its saddle point on the z axis at $z = -1/\sqrt{E}$ where the potential has the value $V = -2\sqrt{E}$. Thus if an atom is in a state of $m = 0$, so that there is no additional centrifugal potential, and if the electron is bound by an energy W, a field given by

$$E = \frac{W^2}{4} \tag{6.35}$$

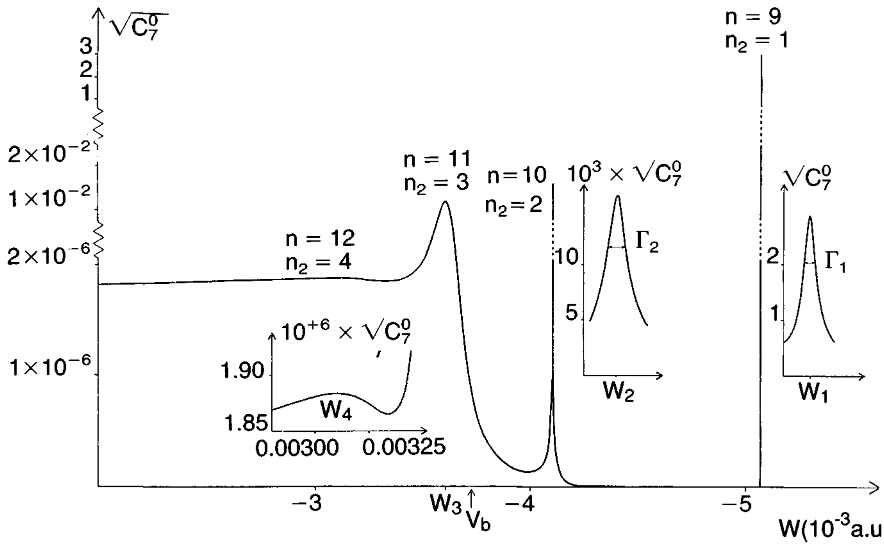

Fig. 6.7 Typical energy dependence of the square root of the density of states $\sqrt{C_{n_1}^m}$ for the states $n_1 = 7$, $|m| = 0$ in the field $E = 1.5 \times 10^{-5}$ au. V_b is the energy of the peak in the potential $V(\eta)$. Each resonance is characterized by the parabolic quantum number n_2 and the width Γ_{n_2} which varies rapidly with n_2. For $W < V_b$ the spectrum exhibits quasi-discrete resonances; for $W > V_b$ the continuum exhibits oscillations which damp out with increasing n_2. The energies and widths of the prominent resonances are

$W_1 = -0.005066$	$\Gamma_1 = 3.7 \times 10^{-16}$ au
$W_2 = -0.004097$	$\Gamma_2 = 6.368 \times 10^{-8}$ au
$W_3 = -0.003601$	$\Gamma_3 = 1.353 \times 10^{-4}$ au
$W_4 = -0.003100$	$\Gamma_4 = 1.637 \times 10^{-3}$ au

(from ref. 3).

is adequate for ionization to occur classically. This field is usually termed the classical field for ionization. If we ignore the Stark shift of a Rydberg state of principal quantum number n and write the binding energy in terms of n we obtain the familiar result for the classical field for ionization in terms of the principal quantum number,

$$E = \frac{1}{16n^4}. \tag{6.36}$$

These results, Eqs. (6.35) and (6.36), are only valid for $m = 0$ states. In higher $|m|$ states there is a $1/(x^2 + y^2)$ centrifugal potential keeping the electron away from the z axis, and the centrifugal barrier raises the threshold field of $m \neq 0$ states.[16] Specifically, for $m \neq 0$ states the fractional increase in the field required for ionization, compared to an $m = 0$ state of the same energy is[16]

$$\frac{\Delta E}{E} = \frac{|m|\sqrt{W}}{\sqrt{2}} = \frac{|m|}{2n}. \tag{6.37}$$

Actually, the classical picture outlined above has two serious defects. First, Eq. (6.36) ignores the Stark shifts, as does Eq.(6.37) if we choose to use the $|m|/2n$ form. Eq. (6.35) and the $|m|\sqrt{W}/\sqrt{2}$ form of Eq. (6.37) do not suffer from this defect. Second, the classical approach ignores the spatial distribution of the wavefunctions. As shown by Fig. 6.2 for H states of the same n and $|m|$, the higher energy, or blue, $n_1 - n_2 \sim n$, Stark states require higher fields to ionize than the lower energy, or red, $n_2 - n_1 \sim n$ states. This point is made quite graphically by Bethe and Salpeter[1] using the data of Rausch von Traubenberg.[17] A simple physical argument for why blue states are harder to ionize than red states is that the electron is located on the side of the atom away from the saddle point in the potential in the blue states, whereas the electron is adjacent to the saddle point in the red states.

In parabolic coordinates the motion in the ξ direction is bounded. Thus for ionization to occur the electron must escape to $\eta = \infty$. Classically, ionization only occurs for energies above the peak V_b of the potential barrier in $V(\eta)$. If we ignore the short range $1/\eta^2$ term in Eq. (6.23b), an approximation good for low m, and set $W = V_b$ we find the required field for ionization to be

$$E = \frac{W^2}{4Z_2}. \tag{6.38}$$

Eq. (6.38) differs from Eq. (6.35) by the factor of Z_2, the effective charge. Consider three low $|m|$ states of the same energy. A blue state of $n_1 - n_2 \approx n$ has $Z_2 \approx 1/n$, a central state has $n_1 = n_2 \sim n/2$ and $Z_2 \approx 1/2$, and a red state has $n_2 - n_1 \sim n$ and $Z_2 \sim 1$. If the three states have the same energy the red one will be most easily ionized.

For the extreme red Stark state of high n Eq. (6.38) reduces to Eq. (6.35) since $Z_2 \approx 1$. For this Stark state the Stark shift increases the binding energy, and for an $m = 0$ state the energy is adequately given using the linear Stark effect as

$$W = -\frac{1}{2n^2} - \frac{3n^2E}{2}. \tag{6.39}$$

Using this energy we find a threshold field

$$E = \frac{1}{9n^4}. \tag{6.40}$$

The numerical factor of 1/9 instead of 1/16 is due to the Stark shift of the level.

For the blue states it is not possible to estimate simply the threshold field. However, blue and red states of the same n and m=0 often have threshold fields differing by a factor of 2.

We have thus far discussed field ionization as occurring only when it is classically allowed, when $E > W^2/4Z_2$, i.e. when the energy is above V_b, the peak of the barrier in $V(\eta)$, as shown in Fig. 6.6. At lower energies tunnelling occurs, and accurate calculations of ionization rates have been made.[2,18–20] As an example, we show in Fig. 6.8 the ionization rates calculated by Bailey *et al.*[20] using

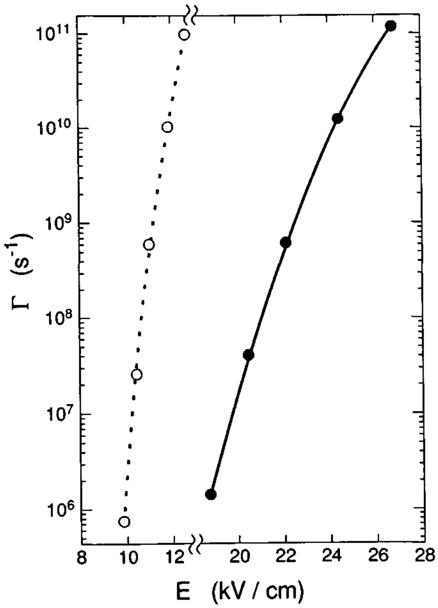

Fig. 6.8 Calculated ionization rates of the (n,n_1,n_2,m) states as a function of field using the values of ref. 20: the (15, 0, 14, 0) red state (○) and the (15, 14, 0, 0) blue state (●).

the WKB method of Rice and Good[18] for the H n=15 extreme red and blue $m = 0$ Stark states over the range of ionization rates from $10^6\ s^{-1}$ to $10^{11}\ s^{-1}$. From Fig. 6.8 it is apparent that the blue states ionize at higher fields than red states of the same n. It is also useful to note that the ionization rates increase rapidly with field, implying that simply computing when ionization is classically allowed is often quite adequate for practical applications.

Using a fast (8 keV) beam of H Koch and Mariani[21] have made precise measurements of the ionization rates of state selected $|m| = 0$ and 1, $n = 30$–40 blue Stark states of H. As can be seen in Fig. 6.9, the measured rates are in quite good agreement with, but typically offset from, the analytic results obtained by Damburg and Kolosov,[22] using the approximation that the field is weak enough that n_2 is very nearly a good quantum number. However, as shown, the measured rates are in excellent agreement with the exact numerical calculations of Damburg and Kolosov,[23] which are in good agreement with the ionization rates calculated by Bailey *et al.*[20] The excellent agreement between theory and experiment suggests that a nonrelativistic treatment is quite adequate. However, Bergeman has calculated that it should be possible to see the effects of spin orbit coupling in field ionization.[24]

The ionization measurements of Koch and Mariani[20] are for blue states of $n_1 \approx n$. Rottke and Welge[25] carried out measurements on the relatively red $|m| = 0$ and 1 states of $n = 18$ and blue states of $n = 19$ verifying the rapid decrease in ionization rate with n_1 shown theoretically in Fig. 6.9. In their measurements

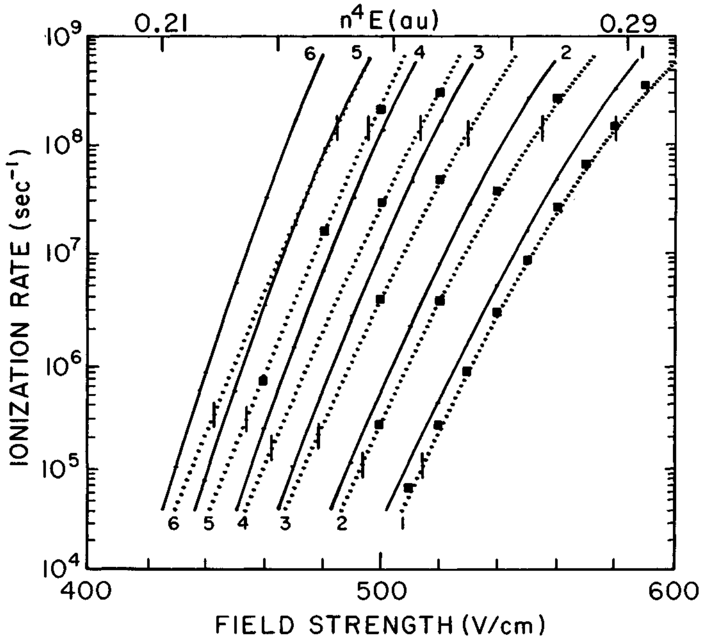

Fig. 6.9 Dotted lines, experimental ionization rate curves for the $(n,n_1,n_2,|m|)$ states 1: (40,39,0,0); 2: (40,38,0,1); 3: (40,38,1,0); 4: (40,37,1,1); 5: (40,37,2,0); 6: (40,36,2,1). The tick marks represent the range of validity of the experiment. Solid line, theoretical curves calculated with Eq. (6) of ref. 22. Squares, numerical theoretical results from ref. 23 (from ref. 21).

they used pulsed laser excitation of a beam of H atoms in a field of 5714 V/cm followed by time resolved detection of the electrons from the ionizing H atoms.

Most of the discussion of field ionization thus far has been focused on states of low $|m|$. For states of high $|m|$ we may no longer neglect the centrifugal potential, and it is not possible to derive a simple expression such as Eq. (6.38). Nonetheless, since the centrifugal term raises the barrier in the potential $V(\eta)$ it is clear that the ionizing field must be higher for high m states. A graphic illustration of this point is provided by Fig. 6.10, a simulation of the ionizing fields of all the $|m| \geq 3$ states of $n = 31$.[26] Measurements of the field ionization profiles of collisionally induced mixtures of the degenerate high $\ell,|m|$ states of the same n are in substantial agreement with the profile predicted in Fig. 6.10.[26]

Nonhydrogenic atoms

Atoms other than H have characteristics essentially similar to those of the H atom in an electric field, but there are important differences due to the presence of the finite sized ionic core. In zero field the presence of the core simply depresses

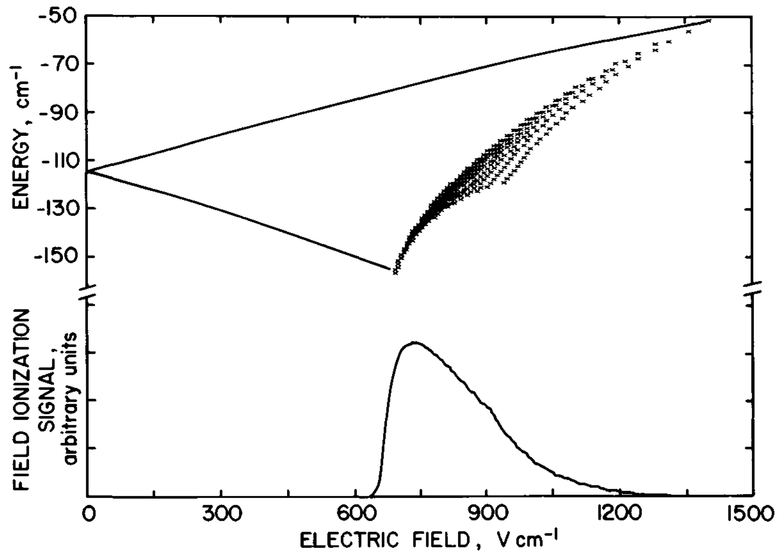

Fig. 6.10 Calculated SFI profile for diabatic ionization of the H like $|m| \geq 3$ states. Top, extreme members of the $n = 31$, $|m| = 3$ Stark manifold. The crosses represent the points at which each $|m| \geq 3$ Stark state achieves an ionization rate of 10^9 s^{-1}. Bottom, calculated SFI profile for diabatic ionization of a mixture containing equal numbers of atoms in each $|m| \geq 3$ Stark level for $n = 31$ at a slew rate of 10^9 V/cm s. (from ref. 26).

the energies, particularly those of the lowest ℓ states. As long as the core is spherically symmetric, it does not alter the spherical symmetry of the problem, and the effect is relatively minor. However, with a finite sized ionic core the wavefunction is no longer separable in parabolic coordinates. As a result the parabolic quantum number n_1, which is a good quantum number in H, is no longer good.

The most important implication of n_1 not being a good quantum number is that blue and red states are coupled by their slight overlap at the core. In the region below the classical ionization limit blue and red states of adjacent n do not cross as they do in H, but exhibit avoided crossings as a result of their being coupled. Above the classical ionization limit blue states, which would be perfectly stable in H, are coupled to degenerate red states, which are unbound, and ionization occurs rapidly compared to radiative decay. It is really an autoionization process in which the blue state is coupled to the red continuum state at the ionic core.

In this chapter we shall treat the Stark effect in nonhydrogenic atoms by methods which, while not the most powerful, provide physical insight. We begin by calculating the energy levels of Na, as an example, in the regime where the field ionization rates are negligible. Ignoring the spin of the Rydberg electron, the Hamiltonian is given by

$$H = -\frac{\nabla^2}{2} + \frac{1}{r} + V_{\rm d}(r) + Ez, \tag{6.41}$$

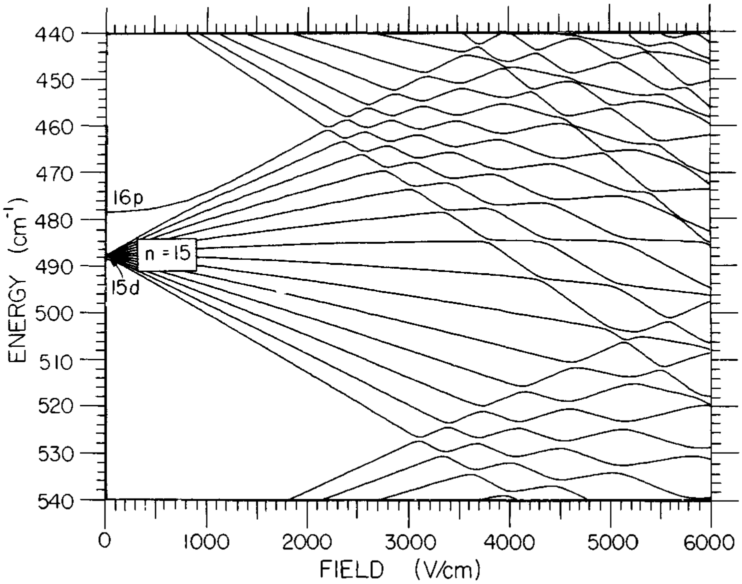

Fig. 6.11 Calculated Na $|m| = 1$ energy levels (from ref. 27).

where $V_d(r)$ is the difference between the Na potential and the coulomb potential, $-1/r$. Consequently, $V_d(r)$ is only nonzero near $r = 0$. A straightforward and effective method of calculating the energies and wavefunctions is direct diagonalization of the Hamiltonian matrix. If we use the Na $n\ell m$ spherical states as basis functions, the Hamiltonian has diagonal elements given by the known energies of the zero field $n\ell m$ states, i.e.

$$\langle n\ell m|H|n\ell m\rangle = \frac{1}{2(n-\delta_\ell)^2}, \tag{6.42}$$

and off diagonal matrix elements $\langle n\ell m|Ez|n'\ell \pm 1m\rangle$ which may be calculated using the Numerov method outlined in Chapter 2. If several n manifolds of ℓ states are included, the eigenvalues of the matrix give the energy levels of the Na atom in the field, and the eigenvectors give the Stark states in terms of the zero field $n\ell m$ states.

In Figs. 6.11 and 6.12 we show the energy levels for the Na states of $n \sim 15$ for $|m| = 1$ and 0, respectively, obtained by direct diagonalization of the Hamiltonian matrix.[27] In zero field the quantum defects 1.35 and 0.85, of the Na s and p states displace them from the high ℓ states and they only exhibit large Stark shifts when they intersect the manifold of Stark states. A second obvious difference between Figs. 6.11 and 6.12 and Fig. 6.2 is that the energy levels of different n do not cross as they do in H. There are avoided crossings, which are clearly visible for the $|m| = 1$ states of Fig. 6.11 and overwhelming for the $m = 0$ states of Fig. 6.12. The energy level diagram for the Na $|m| = 2$ states would not be observably different from the analogous diagram for H on the scale of Figs. 6.11 and 6.12. At zero field

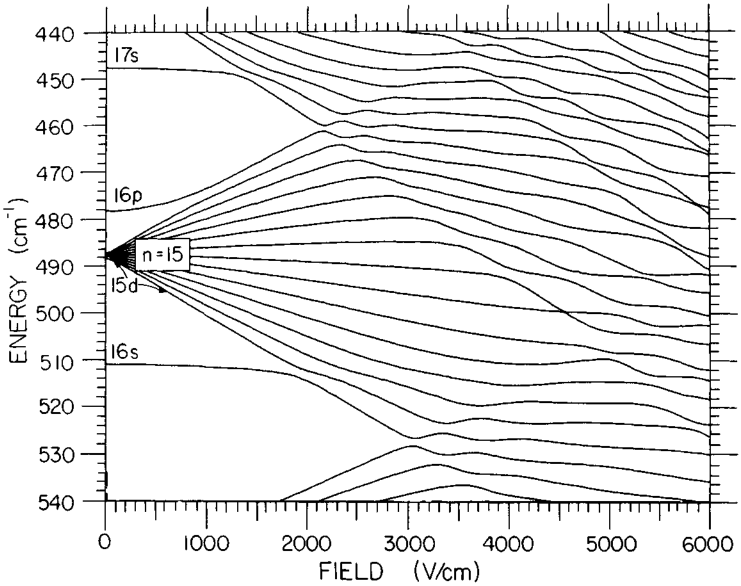

Fig. 6.12 Calculated Na $m = 0$ energy levels (from ref. 27).

the levels would be degenerate, and for $E > 1/3n^5$ the blue and red levels of adjacent n would cross.

A subtle point which is not immediately apparent when inspecting Figs. 6.11 and 6.12 is that the $|m| = 0$ and 1 levels are not interleaved as they are in H. Note that in Fig. 6.12 the 16s state joins the manifold at a field of 2000 V/cm. At fields below 1500 V/cm the $|m| = 0$ and $|m| = 1$ levels of the lower half of the $n = 15$ Stark manifold lie practically on top of one another. This point was demonstrated explicitly by Fabre *et al*[28] who used a diode laser to excite Na atoms from the 3d state to states of the $n = 29$ Stark manifold with high resolution. In Fig. 6.13 we show the result of scanning the wavelength of the diode laser across three high lying pairs of $|m| = 0$ and 1 levels of the $n = 29$ Stark manifold in a field of 20.5 V/cm. The $n = 29$ and $n = 30$ Stark manifolds intersect at a field of 83 V/cm, so the field of 20.5 V/cm corresponds roughly to a field of 550 V/cm in Figs. 6.11 and 6.12, i.e. the 30p state has not yet joined the $n = 29$ Stark manifold. The $|m| = 0$ and 1 states are <200 MHz apart while the adjacent pairs of Stark levels are ~2 GHz apart. In H the $|m| = 0$ and 1 levels would be 1 GHz apart. The $|m| = 0$ levels lie above the $|m| = 1$ levels, i.e. they are further displaced from the center of the manifold, or the zero field $n = 29$ high ℓ levels.

Fabre *et al.*[28] used a projection operator technique to describe the Stark shifts at fields below where low ℓ states of large quantum defects join the manifold. A less formal explanation is as follows. If, for example, the s and p states are excluded, as in Fig. 6.13 below 800 V/cm, effectively only the nearly degenerate $\ell \geq 2$ states are coupled by the electric field. The only differences among the $|m| = 0,1$, and 2 manifolds occur in the angular parts of the matrix element, i.e.[1]

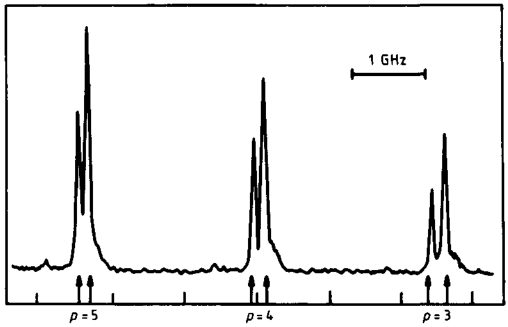

Fig. 6.13 Part of the excitation spectrum of the Na $n = 29$ Stark levels from the 3d state in an electrostatic field of 20.5 V/cm corresponding to n_1 values of $n - 3$, $n - 4$, and $n - 5$. The energy splitting between $m = 0$ (highest energy line in the doublets) and $|m| = 1$ states is of the order of 180 MHz. The arrows indicate theoretical positions of energy levels obtained by a numerical diagonalization of the Stark Hamiltonian (from ref. 28).

$$\langle n\ell m|\cos\theta|n\ell + 1m\rangle = \sqrt{\frac{(\ell+1)^2 - m^2}{(2\ell+3)(2\ell+1)}}. \tag{6.43}$$

Only in a few low ℓ states is the m dependence significant, and as a result the $|m| = 0,1$, and 2 energy levels are almost degenerate. However, due to the low ℓ states, the $|m| = 0$ levels are displaced more than the $|m| = 1$ levels, which are displaced insignificantly more than the $|m| = 2$ levels.

Reexamining Figs. 6.11 and 6.12, the avoided crossings of the blue and red states of adjacent n are evident. When the avoided crossings are large they can be measured optically as was done by Zimmerman *et al.*[27] They excited Li atoms in a beam from the 2s state to the 2p state and then to the 3s state by two pulsed dye lasers at fixed wavelengths. The wavelength of a third laser was scanned to drive atoms from the 3s state to Rydberg Stark states in the presence of a static field. The transition to the Rydberg state was detected by applying a high field pulse to the atoms subsequent to the laser pulses. The ions were detected with a particle multiplier as the wavelength of the third laser was scanned. In Fig. 6.14 we show their observation of the avoided crossing of the (18,16,0,1) and (19,1,16,1) states. The experimental picture of Fig. 6.14 is built up by scanning the third laser over the energy of the avoided crossing for several fields near the avoided crossing field of 943 V/cm. As shown by Fig. 6.14, the observed energies of the levels match those calculated by matrix diagonalization. It is also interesting to note that the oscillator strength to the upper state vanishes at the anticrossing. At the anticrossing the eigenstates have the form $(1/\sqrt{2})$ $(\Psi_A \pm \Psi_B)$ where Ψ_A and Ψ_B are the eigenstates away from the crossing. As shown by Fig. 6.14, away from the avoided crossing both states apparently have similar amounts of p character, and at the crossing the upper state loses it all to the lower state.

The method used by Zimmerman *et al.*[27] to measure the Li avoided crossing shown in Fig. 6.14 requires that the resolution of the laser be finer than the size of

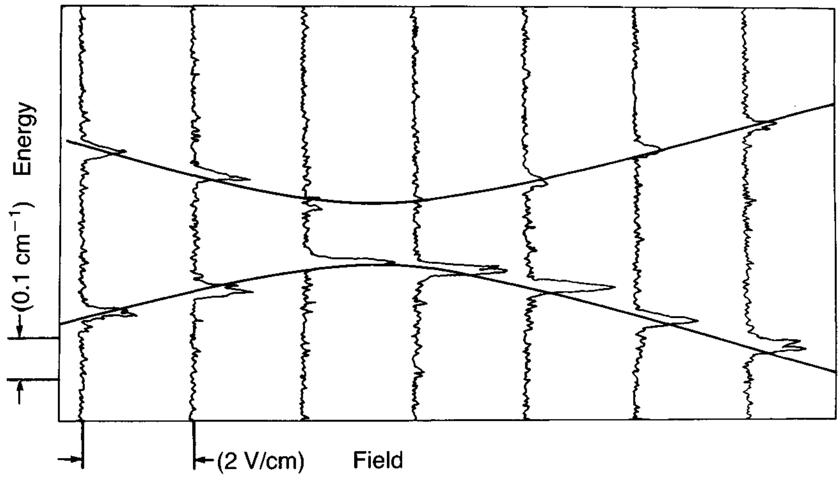

Fig. 6.14 Level anticrossing in Li. Intersection of the $(n,n_1,n_2,|m|)$ states (18,16,0,1) and (19,1,16,1) at 321.5 cm^{-1} and 943 V/cm. The calculated level structure is superimposed on the data. Note the vanishing oscillator strength to the upper level at the avoided crossing (from ref. 27).

the avoided crossing. An alternative approach used by Stoneman *et al.*[29] bypasses this requirement. They observed the narrow anticrossings between the K $(n+2)$s states, which have a quantum defect of 2.18, and the low lying members of the n Stark manifold, which is composed of $\ell \geq 3$ states. They excited atoms in a beam from the 4s state to the 4p state and then to the ns state with two pulsed dye lasers with linewidths of 1 cm^{-1}. Two to three microseconds after the laser excitation the atoms were selectively field ionized, and only atoms which had undergone black body radiation induced transitions to the higher lying $(n+2)$p states were detected. The $(n+2)$p signal was recorded as the static field was slowly scanned through the avoided crossing field over many shots of the laser. Away from the avoided crossing only the $(n+2)$s state was excited, and it could undergo black body radiation induced transitions to the $(n+2)$p state. The n Stark state, composed of $\ell \geq 3$ states, cannot be excited from the 4p state or to the $(n+2)$p state. At the avoided crossing the eigenstates are 50–50 mixtures of the two states, and both are excited from the 4p state, with the result that the same number of Rydberg atoms is always excited. However, each of these two states has half the $(n+2)$s–$(n+2)$p black body radiation induced transition rate, and the observed $(n+2)$p signal decreases. An example of their signals, the avoided crossing signals of the K 20s state with the lowest lying $|m| = 0$ and 1 states of the $n = 18$ Stark manifold, is shown in Fig. 6.15.[29] First we note the decrease in the 20p ion signal at each of the avoided crossings, as described above. Second we note that the $|m| = 0$ and 1 Stark states are nearly degenerate. If they were interleaved as in H, the two avoided crossings would be separated by 50 V/cm not 5 V/cm as they are in Fig. 6.15. We have again encountered the noninterleaving of the energy

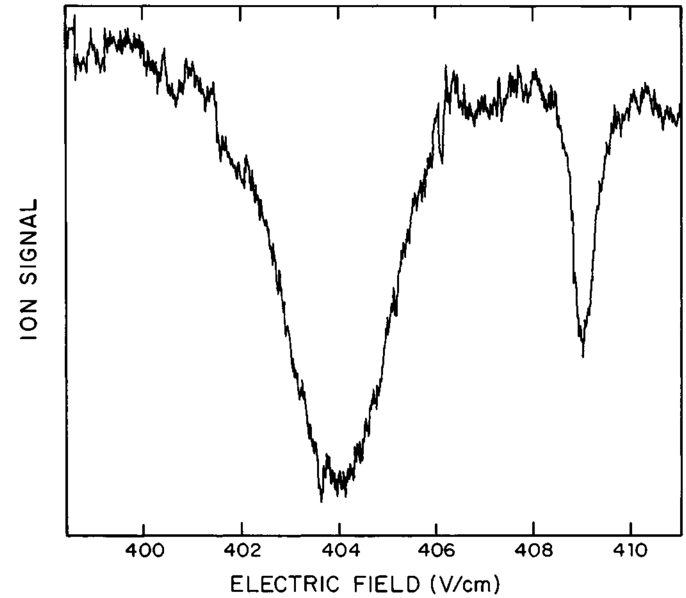

Fig. 6.15 Anticrossing signal from the avoided crossing of the K 20s state with the lowest energy $n = 18$ Stark manifold state. The left-hand peak corresponds to the $m = 0$ anticrossing, and the right-hand peak to the $|m| = 1$ anticrossing (from ref. 29).

levels of different m, as shown in Fig. 6.13. The spin orbit interaction, which we have ignored, is responsible for the avoided crossing between the 20s state and the $|m| = 1$ Stark state. Diagonalizing the Hamiltonian matrix, including the spin orbit interaction, yields 450 and 47 MHz for the widths of the $|m| = 0$ and 1 anticrossings of Fig. 6.16, in reasonably good agreement with the measured values of 510 MHz and 80 MHz.[29] In the heavier alkali atoms the spin orbit interaction plays an even more dominant role.[27]

When the quantum defects are large and $|m|$ is low enough that several states have large quantum defects, matrix diagonalization using the $n\ell m$ states as basis functions is probably the most efficient approach. However, if the contributing quantum defects are small, a different approach to Eq. (6.41), suggested by Komarov *et al.*[30] is useful. They suggested using the hydrogenic parabolic states as a basis instead of the zero field $n\ell m$ states, in which case the Hamiltonian of Eq. (6.41) has diagonal matrix elements

$$\langle nn_1n_2m|H|nn_1n_2m\rangle = \\ -\frac{1}{2n^2} + \frac{3}{2}n(n_1 - n_2)E + O(E^2) + \langle nn_1n_2m|V_d(r)|nn_1n_2m\rangle, \quad (6.44)$$

and nonzero off diagonal elements,

$$\langle nn_1n_2m|H|n'n_1'n_2'm\rangle = \langle nn_1n_2m|V_d(r)|n'n_1'n_2'm\rangle. \quad (6.45)$$

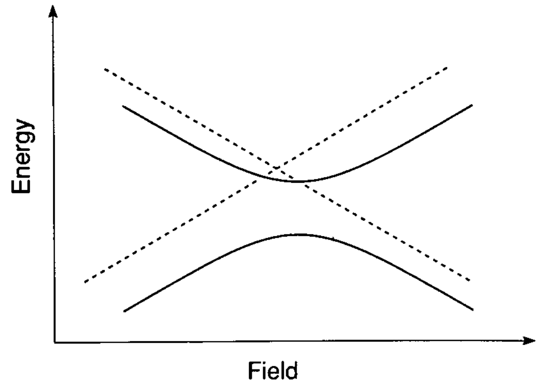

Fig. 6.16 Hydrogenic Stark energy levels (- -) before considering the effect of the core perturbation $V_d(r)$. The alkali levels after incorporating the diagonal and off diagonal matrix elements of $V_d(r)$ (——). The diagonal elements shift the levels and the off diagonal elements lift the degeneracy at the level crossing. Note that the different shifts of the levels move the avoided crossing from the field of the hydrogenic crossing.

The diagonal elements are the hydrogenic energies plus a small energy shift, and the off diagonal elements are the couplings between levels. In Fig. 6.16 we show two hydrogenic levels as dotted lines and the non hydrogenic levels as solid lines. The shifts of the nonhydrogenic levels from their hydrogenic counterparts are due to the diagonal matrix elements of $V_d(r)$ and the avoided crossings to the off diagonal matrix elements of $V_d(r)$.

It is straightforward to evaluate the matrix elements of $V_d(r)$ using the known transformation between parabolic and spherical states given in Eqs. (6.18) and (6.19). If we expand the parabolic states in terms of hydrogenic spherical states, the matrix elements of $V_d(r)$ may be written as[30]

$$\langle nn_1n_2m|V_d(r)|n'n_1'n_2'm\rangle = \sum_\ell \langle nn_1n_2m|n\ell m\rangle\langle n\ell m|V\ (r)|n'\ell m\rangle\langle n'\ell m|n'n'm\rangle, \tag{6.46}$$

where we have taken advantage of the spherical symmetry of $V_d(r)$ to require the same ℓ on both sides of the matrix element. We recall that the core interaction causes the depression of the energy levels below their H analogues. Thus,

$$\langle n\ell m|V_d(r)|n\ell m\rangle = \frac{-\delta_\ell}{n^3}. \tag{6.47}$$

This expression may be generalized by realizing that $V_d(r)$ is only nonzero for small r, where the dependence on n is through the normalization factor $n^{-3/2}$ Accordingly Eq. (6.47) can be generalized to[30]

$$\langle n\ell m|V_d(r)|n'\ell m\rangle = \frac{-\delta_\ell}{\sqrt{n^3n'^3}}. \tag{6.48}$$

Thus,

$$\langle nn_1n_2m|V_d|n'n_1'n_2'm\rangle = \sum_\ell \langle nn_1n_2m|n\ell m\rangle \frac{-\delta_\ell}{\sqrt{n^3n'^3}} \langle n'\ell m|n'n_1'n_2'm\rangle. \quad (6.49)$$

Since $|\langle nn_1n_2m|n\ell m\rangle|^2$ is the $n\ell m$ fraction of the nn_1n_2m state, it is evident that the depression of the nonhydrogenic Stark levels below the hydrogenic value, given by the diagonal matrix elements of $V_d(r)$, is simply the amount of each low ℓ state it possesses multiplied by its quantum defect. The nonhydrogenic energy must be lower than the hydrogenic energy since $\delta_\ell > 0$ for all ℓ.

At the field where the diagonal energies of two levels, given by Eq. (6.43), are equal, the off diagonal matrix element lifts the degeneracy. At the avoided crossing the energy levels are split by

$$\Delta W = 2|\langle nn_1n_2m|V_d(r)|n'n_1'n_2'm\rangle|, \quad (6.50)$$

and the eigenstates are symmetric and antisymmetric combinations of the two Stark states. Eq. (6.49) shows how the avoided crossings scale with n. Averaged over ℓ, $\langle nn_1n_2m|n\ell m\rangle^2$ must be equal to $1/(n-m)$. Thus, for low $|m| \ll n$ and $n \sim n'$, the matrix element of Eq. (6.49) is proportional to $1/n^4$, and the avoided crossings exhibit a $1/n^4$ scaling.

Ionization of nonhydrogenic atoms

A nonhydrogenic atom, Na for example, differs from H not only in its energy level spectrum but also in how it is ionized by a field. In particular, in a nonhydrogenic atom there are two forms of field ionization. The first is exactly like H. A state of approximate quantum number n_1 can itself ionize, keeping the same value of n_1, in a field E if its energy W is enough for the electron to surmount the potential barrier V_b in $V(\eta)$. Rates for this form of ionization increase rapidly in the vicinity of the threshold for classical ionization, just as in H. In this form of ionization the blue states require higher fields for ionization than red states of the same energy.

The second form of ionization is similar to autoionization.[31] In a nonhydrogenic atom n_1 is not a good quantum number and bound states of high n_1 are coupled to Stark continua of low n_1. This form of ionization applies to those states other than the reddest Stark states. The extreme red Stark states have $n_1 = 0$ and ionize, as do the red H states, at the classical ionization limit given by Eq. (6.35), modified by Eq. (6.37) for $m \neq 0$ states. This point has been demonstrated explicitly by Littman *et al*,[32] who measured the time resolved ionization of Na $|m| = 2$ states subsequent to pulsed laser excitation. Their measured rates are in good agreement with the rates obtained by extrapolation of the rates of Bailey *et al.*[20] shown in Fig. 6.8.

In H it is possible to have $n_1 \geq 1$ states which are quite stable against ionization even though energetically degenerate states of lower n_1 have been converted to continua. In other atoms these blue $n_1 \geq 1$ states are coupled by the finite core to the red continua and autoionize. A simple physical picture is that an electron in a blue orbit comes near the core once on each orbit and has a finite probability of being scattered from the stable blue orbit into the degenerate red continuum. A significant difference between this form of ionization and hydrogenic ionization is that its rate is not exponentially dependent on the field.

The existence of the second form of ionization means that there is a large regime of field and energy in which nonhydrogenic atoms have states which decay more rapidly by ionization than by radiative decay, but are still spectroscopically narrow. In H all states have ionization rates which are exponentially dependent on the field so the analogous range is much smaller. The fact that there exist sharp but ionizing levels above the classical ionization limit but below the hydrogenic ionization limit was first observed by Jacquinot *et al.* in Rb.[33] It was most clearly shown by Littman *et al.* in Li.[31] They excited a beam of Li atoms in static fields by two fixed frequency lasers from the ground state 2s via the 2p state to the 3s state. They then scanned the wavelength of a third laser to drive transitions from the 3s state to Rydberg states having np character. The excitation to a Rydberg state was detected by field ionization of the atom and subsequent detection of the ion with a particle multiplier. Spectra were recorded at many static fields, as shown by Fig. 6.17, which is composed of spectra recorded with the third laser polarized perpendicular to the static field so as to excite only $|m| = 1$ states. The spectra of Fig. 6.17 were recorded in two different ways. The spectra of Fig. 6.17(a) were recorded by applying a field ionization pulse 3 μs after laser excitation and only detecting the ions which result from ionization by the pulse, not spontaneous ionization. The resulting spectra are then composed of states which are stable for at least 3 μs against field ionization, or radiative decay, which is longer than 3 μs for the states of Fig. 6.17. It is quite apparent that there are no stable states above the classical ionization limit.

On the other hand, if no field ionization pulse is applied and the ions from atoms which ionize in the first 3 μs after the laser excitation are detected, the result is Fig. 6.17(c). The spectra begin at the classical limit and extend to much higher static fields. What is most important to note is that levels appear over a field range of typically a factor of 2 with no apparent change in width, a phenomenon which does not occur in H. Fig. 6.17(b) shows the calculated H $|m| = 1$ energy levels. Inspecting Fig. 6.17(c) it is easy to see that the levels disappear from the experimental spectrum when their decay rates exceed the laser linewidth by a large margin.

We may develop a more quantitative understanding of nonhydrogenic ionization by starting with the hydrogenic Stark states and adding the core perturbation $V_d(r)$, just as we calculated the sizes of the avoided level crossings. The level crossings described earlier are the result of core couplings between the bound

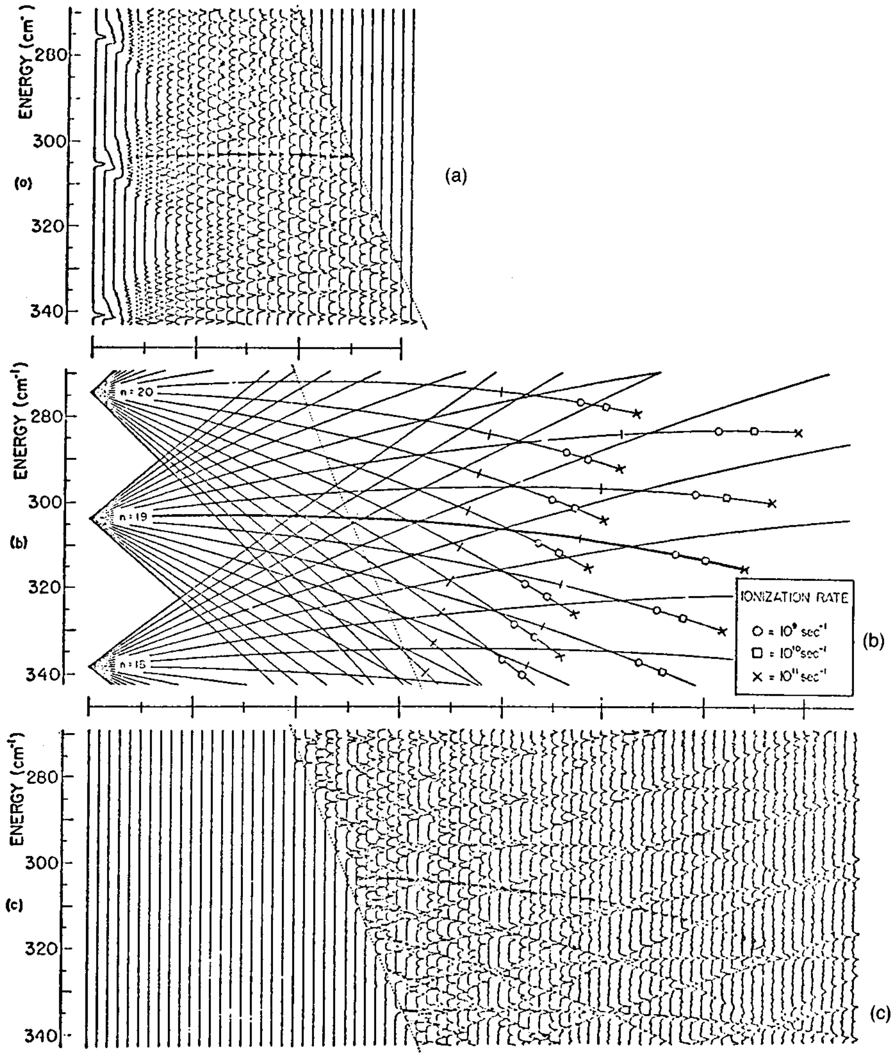

Fig. 6.17 Tunnelling and saddle point ionization in Li. (a) Experimental map of the energy levels of Li $|m| = 1$ states in a static field. The horizontal peaks arise from ions collected after laser excitation. Energy is measured relative to the one-electron ionization limit. Disappearance of a level with increasing field indicates that the ionization rates exceed $3 \times 10^5\ s^{-1}$. The dotted line is the classical ionization limit given by Eqs. (6.35) and (6.36). One state has been emphasized by shading. (b) Energy levels for H ($n = 18$–20, $|m| = 1$) according to fourth order perturbation theory. Levels from nearby terms are omitted for clarity. Symbols used to denote the ionization rate are defined in the key. The tick mark indicates the field where the ionization rate equals the spontaneous radiative rate. (c) Experimental map as in (a) except that the collection method is sensitive only to states whose ionization rate exceeds $3 \times 10^5\ s^{-1}$. At high fields, the levels broaden into the continuum in agreement with tunnelling theory for H (from ref. 32).

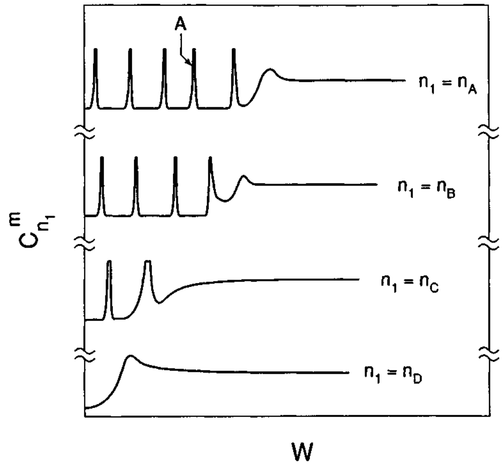

Fig. 6.18 The density of states $C_{n_1}^m$ for four different values of the quantum number n_1: $n_A > n_B > n_C > n_D$. For each value of n_1 $C_{n_1}^m$ is composed of narrow resonances up the $W = V_b$, the energy of the saddle point in $V(\eta)$. Above $W = V_b$ $C_{n_1}^m$ is continuous as shown. The bound state A in $n_1 = n_A$ can ionize into the $n_1 = n_C$ and n_D continua in a nonhydrogenic atom. There is often a peak in $C_{n_1}^m$ at the onset of the continuous spectrum, as shown.

nn_1n_2nm and $n'n_1'n_2'm$ states, couplings between bound levels of different n_1. The same coupling between a hydrogenically bound n_1m state and a continuum state of lower n_1 results in ionization.

Ignoring for the moment coupling between nominally bound states of different n_1, the ionization rate of the state nn_1m to the $\varepsilon n_1'm$ continua is given by Fermi's Golden Rule,

$$\Gamma = 2\pi \sum_{n_1} |\langle nn_1m|V_d(r)|\varepsilon n_1'm\rangle|^2. \tag{6.51}$$

The summation extends over all values of n_1' which are continua at the energy and field in question.

Since $V_d(r)$ is only nonzero near $r = 0$ the matrix element of Eq. (6.51) reflects the amplitude of the wavefunction of the continuum wave at $r \sim 0$. Specifically, the squared matrix element is proportional to $C_{n_1}^m$, the density of states defined earlier and plotted in Fig. 6.18. From the plots of Fig. 6.18 it is apparent that the ionization rate into a continuum substantially above threshold is energy independent. However, as shown in Fig. 6.18, there is often a peak in the density of continuum states just at the threshold for ionization, substantially increasing the ionization rate for a degenerate blue state of larger n_1. This phenomenon has been observed experimentally by Littman *et al.*[32] who observed a local increase in the ionization rate of the Na (12,6,3,2) Stark state where it crosses the 14,0,11,2 state, at a field of 15.6 kV/cm, as shown by Fig. 6.19. In this field the energy of the

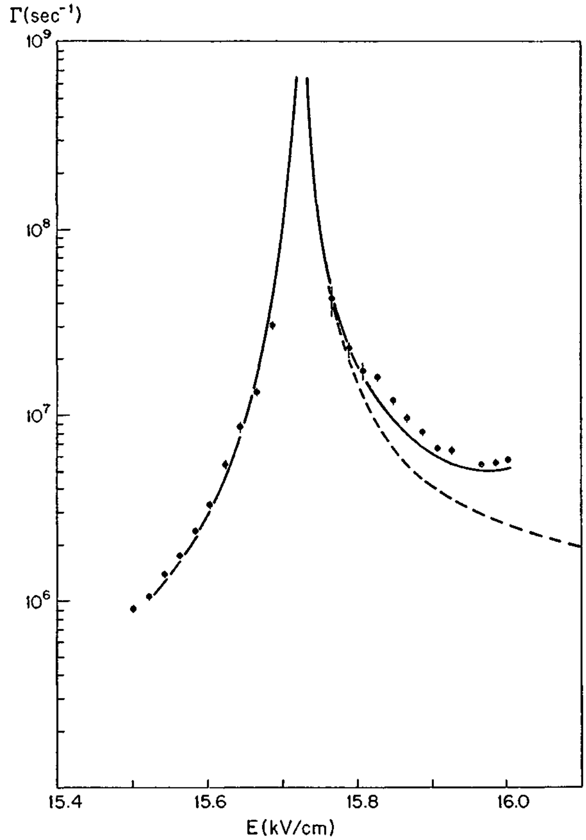

Fig. 6.19 Ionization rates for the (12,6,3,2) level of Na in the region of the crossing with the rapidly ionizing (14,0,11,2) level. The levels are specified as $(n,n_1,n_2,|m|)$. The solid line is a theoretical curve. The dashed line is calculated neglecting another avoided level crossing at 17.3 kV/cm (from ref. 31).

(14,0,11,2) state is just below the top of the barrier in $V(\eta)$ and its ionizing rate is $10^{10}\ \mathrm{s}^{-1}$. With this low an ionization rate the (14,0,11,2) wavefunction still has a large amplitude at the core, and the autoionization rate of blue states into it is at its highest.

Finally, it is useful to obtain an estimate of the typical ionization rates in the region between the classical ionization limit and the hydrogenic limit using Eq. (6.51). Each term in the sum of Eq. (6.51) can be written as

$$|\langle nn_1m|V_d(r)|\varepsilon n_1'm\rangle|^2 = |\langle nn_1m|n\ell m\rangle\langle n\ell m|V_d|\varepsilon\ell m\rangle\langle\varepsilon\ell m|\varepsilon n_1'm\rangle|^2 \qquad (6.52)$$

The maximum value of n_1' in Eq. (6.51) is the value of n_1 for which $Z_2 = W^2/4E$, or $Z_1 = 1 - Z_2 = 1 - W^2/4E$. As shown by Harmin,[14] a continuum form of Eq. (6.19), valid for small radii, is

$$\langle\varepsilon\ell m|\varepsilon n_1'm\rangle = (-1)^\ell P_{\ell m}(Z_1 - Z_2). \qquad (6.53)$$

We are now in a position to evaluate Eq. (6.51) using the explicit form of Eq. (6.52) and replacing the sum over n_1 by an integral over $Z_1 - Z_2$. The first term of the matrix element of Eq. (6.52) is given by Eqs. (6.11) and (6.19), the second one by the continuum modification of Eq. (6.48), and the third by Eq. (6.53). In addition to these substitutions we convert the summation of Eq. (6.51) to an integral over $Z_1 - Z_2$. According to Eq. (6.38), the classical ionization limit occurs at $E = W^2/4Z_2$. Consequently, the contributing range of Z_2 is from $W^2/4E$ to 1. Written in terms of $Z_1 - Z_2$ the integral corresponding to the ionization rate of Eq. (6.51) is given by

$$\Gamma = 2\pi P_{\ell m}^2 \left[\left(\frac{2n_1}{n}\right) - 1\right] \frac{\sin^2 \delta_\ell'}{n^3} \int_{-1}^{1-W^2/2E} P_{\ell m}^2 (Z_1 - Z_2) \mathrm{d}(Z_1 - Z_2), \tag{6.54}$$

where δ_ℓ' is the magnitude of the difference between δ_ℓ and the nearest integer. For example, $\delta_\ell = 0.85$ implies $\delta_\ell' = 0.15$. The first term in Eq. (6.54) reflects the amount of ℓ character in the initial nn_1m state, the second term is the coupling to the ℓ part of the continuum, and the third, the integral, reflects how much of the continuum is available having ℓ character. The value of the integral ranges from zero at energies just above the classical ionization limit to one when $W = -\sqrt{2E}$.

Approximating $\langle nn_1n_2m|n\ell m\rangle^2$ by $1/n$ and the integral of Eq. (6.54) by 1/2 we find the approximate result

$$\Gamma \sim \pi \frac{\sin^2 \delta_\ell'}{n^4} \tag{6.55}$$

Observations to date indicate that Na $|m| = 0, 1$, and 2 Stark states of $n \approx 20$ above the classical ionization limit have widths of ~30–90 GHz, 1–3 GHz, and 100 MHz,[34] in rough accord with Eq. (6.55).

In estimating ionization rates we have ignored all interactions between bound states of high n_1. These interactions also lead to local variations in the autoionization rates which can be quite striking and can be understood in a straightforward way. For a moment let us disregard the coupling to the continuum and focus on two nominally bound Stark states which have an avoided crossing. As we have already discussed, at the avoided crossing the eigenstates are linear combinations of the two Stark states. If away from the avoided crossing one of the two Stark levels is rapidly ionizing and the other slowly ionizing, at the crossing the ionization rates of the two eigenstates will most likely be about the same and half the rate of the rapidly ionizing state.

A more interesting case is one in which the two Stark states have equal autoionization rates. If the coupling is to the same continuum in both cases, then at the avoided crossing the coupling will vanish in one of the eigenstates and double in the other one. The phenomenon is basically the same as the vanishing of the oscillator strength to the upper level at the avoided crossing of Fig. 6.14. If the two Stark states do not have precisely the same ionization rates away from the avoided crossing, one of the ionization rates cannot vanish at the avoided

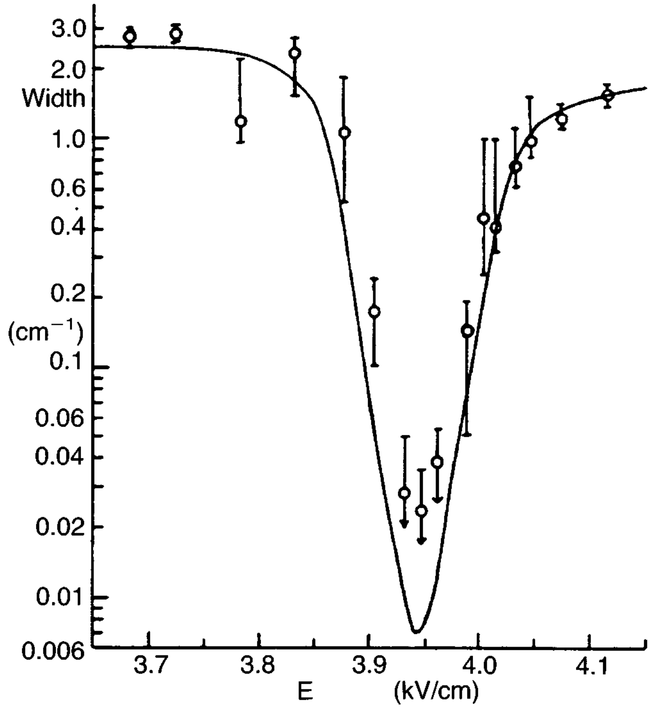

Fig. 6.20 Ionization width vs electric field for the Na (20,19,0,0) level near its crossing with the (21,17,3,0) level from experiment (data points) and from WKB–quantum defect theory (solid line). The levels are specified as $(n,n_1,n_2,|m|)$ Because the lineshapes are quite asymmetric (except for very narrow lines), the width in this figure is taken to be the FWHM of the dominant feature corresponding to the (20,19,0,0) level in the photoionization cross section. For the narrowest line, experimental widths are limited by the 0.7 GHz laser linewidth. Error limits are asymmetric because of the peculiar line shapes and because of uncertainties due to the overlapping $|m| = 1$ resonance (from ref. 37).

crossing. This phenomenon was first observed by Feneuille *et al.*[35] and discussed further by Luc–Koenig and LeCompte.[36] However, it is shown most clearly in the data of Liu *et al.*[37] who measured the width of the Na (20,19,0,0) state in the vicinity of its intersection with the (21,17,3,0) state at 3.95 kV/cm. As shown by Fig. 6.20, the width of the state decreases from 2 cm^{-1} away from the avoided crossing to 0.02 cm^{-1} at the avoided crossing. Correspondingly, the width of the (21,17,3,0) state approximately doubles at the avoided crossing.

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